

Introduction to Bioinformatics with Oxford Nanopore Sequencing

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Quy trình tổng quát



Nội dung buổi 4

1. Align trình tự FASTQ vào hệ gen tham chiếu
2. Convert SAM -> BAM
3. Hiểu kết quả alignment
4. Tính coverage
5. Xem kết quả alignment sử dụng IGV

PHẦN 1

Alignment

Alignment sử dụng Minimap2

```
conda activate ont
```

```
minimap2 -ax map-ont \  
-R '@RG\tID:HG002\tSM:HG002\tPL:ONT' \  
data/reference/GRCh38.chr20.fa \  
data/fastq/HG002.chr20.fastq.gz \  
> results/02_alignment/HG002.chr20.sam
```

```
# chuyển sang BAM và sort luôn
```

```
minimap2 -ax map-ont \  
-R '@RG\tID:HG002\tSM:HG002\tPL:ONT' \  
data/reference/GRCh38.chr20.fa\  
data/fastq/HG002.chr20.fastq.gz | samtools sort -o  
results/02_alignment/HG002.chr20.bam
```

```
# index bam
```

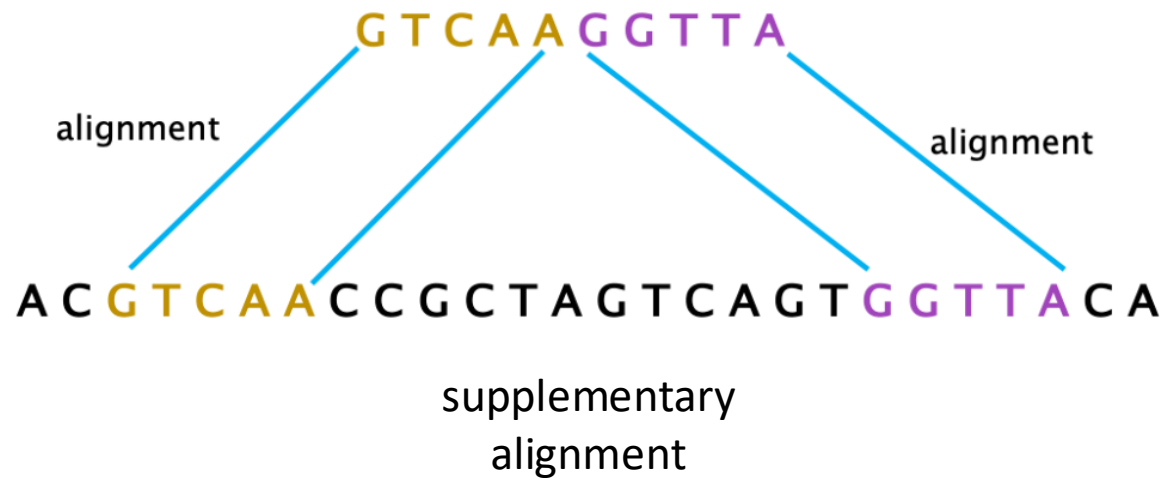
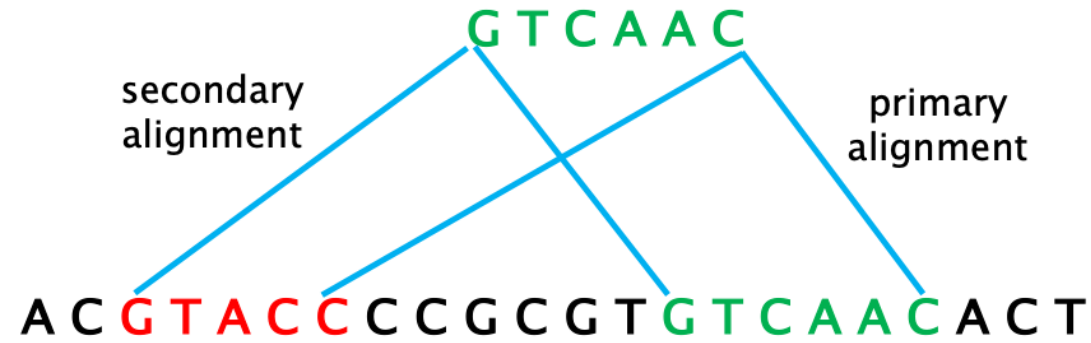
```
samtools index results/02_alignment/HG002.chr20.bam
```

Kiểm tra kết quả alignment

```
samtools flagstats results/02_alignment/HG002.chr20.bam
```

```
> samtools flagstats results/02_alignment/HG002.chr20.bam
2538 + 0 in total (QC-passed reads + QC-failed reads)
1727 + 0 primary
429 + 0 secondary
382 + 0 supplementary
0 + 0 duplicates
0 + 0 primary duplicates
2538 + 0 mapped (100.00% : N/A)
1727 + 0 primary mapped (100.00% : N/A)
0 + 0 paired in sequencing
0 + 0 read1
0 + 0 read2
0 + 0 properly paired (N/A : N/A)
0 + 0 with itself and mate mapped
0 + 0 singletons (N/A : N/A)
0 + 0 with mate mapped to a different chr
0 + 0 with mate mapped to a different chr (mapQ>=5)
```

Primary, secondary, and supplementary



Cấu trúc của SAM/BAM

```
@HD VN:1.6 S0:unsorted G0:query
@SQ SN:chr20 LN:64444167
@PG ID:minimap2 PN:minimap2 VN:2.31-r1302 CL:minimap2
@PG ID:samtools PN:samtools PP:minimap2 VN:1.24 CL:samtools
d4ff2a19-9db8-4bea-8f57-4dbc1a87648a 16 chr20 61981 1
d4ff2a19-9db8-4bea-8f57-4dbc1a87648a 272 chr20 60256 0
b66d7782-f216-43a2-94c9-16cd187996e5 0 chr20 60337 18
b66d7782-f216-43a2-94c9-16cd187996e5 256 chr20 60231 0
0 0 61928 0
0 0 31889969
0 0 60210 6
```

Có 11 cột trong file SAM/BAM

HWI-ST330:304:H045HADXX:20934#0

16

chr1

60023

50

100M

*

0

0

CCACTATGTTTTTCGATAAAAAGCTTAATAAAT

?????BBBBBDBDB=?FFECFACCFHHH>09C

↑

QNAME

↑

FLAG

↑

RNAME

↑

POS

↑

MAPQ

↑

CIGAR

HWI-ST330:304:H045HADXX:20934#0

16

chr1

60023

50

100M

HWI-ST330:304:H045HADXX:20934#0

16

chr1

60023

50

100M

*

0

0

CCACTATGTTTTTCGATAAAAAGCTTAATAAAT

?????BBBBBDBDB=?FFECFACCFHHH>09C

↑

SEQ

↑

QUAL

↑

MRNM

↑

MPOS

↑

ISIZE

*

0

0

Unsorted vs. Sorted BAM

```
@HD      VN:1.6  S0:unsorted      G0:query
@SQ      SN:chr20      LN:64444167
@PG      ID:minimap2    PN:minimap2    VN:2.31-r1302  CL:minimap2 -
@PG      ID:samtools    PN:samtools    PP:minimap2    VN:1.24 CL:sa
d4ff2a19-9db8-4bea-8f57-4dbc1a87648a  16      chr20    61981    1
d4ff2a19-9db8-4bea-8f57-4dbc1a87648a  272     chr20    60256    0
b66d7782-f216-43a2-94c9-16cd187996e5  0       chr20    60337    18
b66d7782-f216-43a2-94c9-16cd187996e5  256     chr20    60231    0
b66d7782-f216-43a2-94c9-16cd187996e5  256     chr20    61928    0
b66d7782-f216-43a2-94c9-16cd187996e5  2064    chr20    31889969
389a34f2-cca8-4afd-9337-4ecae4d619ad  16      chr20    60210    6
```

Unsorted

```
@HD      VN:1.6  S0:coordinate
@SQ      SN:chr20      LN:64444167
@PG      ID:minimap2    PN:minimap2    VN:2.31-r1302  CL:minimap2 -a
@PG      ID:samtools    PN:samtools    PP:minimap2    VN:1.24 CL:sam
@PG      ID:samtools.1  PN:samtools    PP:samtools    VN:1.24 CL:sam
13638a75-308a-404d-b9c7-36cbe43ec75f  2048    chr20    60001    60
cf03ee65-e1ba-405b-bd0e-2c1ccb1bfb7b  2048    chr20    60001    60
d47160cc-0228-45c3-96c4-cf694f28cedb  2048    chr20    60001    60
7c3bee3f-6677-45a2-b3a6-a99ef6602d44  2048    chr20    60001    60
59fb0349-7102-4c4b-8150-f9eff3810030  2048    chr20    60001    60
70974e06-5e08-41d0-9e3d-dc16fea08743  2048    chr20    60001    60
a562ae0b-a6c0-450a-9ba3-c61b476e3a09  2048    chr20    60001    60
21be91d5-e04f-42dd-8d90-40f062af7986  2048    chr20    60001    60
```

Sorted

Hiểu FLAG trong SAM/BAM

- FLAG = một số nguyên mô tả trạng thái của mỗi alignment record
- Một FLAG có thể chứa **nhiều trạng thái cùng lúc**, vì các giá trị được cộng lại theo **bitwise flag**.
- FLAG 272 = 256 + 16
secondary (256) + reverse strand (16)

Hex	Decimal	Ý nghĩa	Liên quan đến ONT
0x1	1	Read is paired	✗
0x2	2	Properly paired	✗
0x4	4	Read is unmapped	✓
0x8	8	Mate is unmapped	✗
0x10	16	Read aligned to reverse strand	✓
0x20	32	Mate aligned to reverse strand	✗
0x40	64	First read in pair	✗
0x80	128	Second read in pair	✗
0x100	256	Secondary alignment	✓
0x200	512	Failed vendor/platform QC	Sometimes
0x400	1024	Marked as duplicate	Sometimes
0x800	2048	Supplementary alignment	✓

Thống kê theo flag?

```
samtools view results/02_alignment/HG002.chr20.bam \
| cut -f2 \
| sort -n \
| uniq -c \
| xargs -n2 sh -c 'echo "$1\t$(samtools flags "$2")"' _
```

927	0x0	0	Primary mapped
800	0x10	16	REVERSE
166	0x100	256	SECONDARY
263	0x110	272	REVERSE, SECONDARY
174	0x800	2048	SUPPLEMENTARY
208	0x810	2064	REVERSE, SUPPLEMENTARY

```
> samtools flagstats results/02_alignment/HG002.chr20.bam
2538 + 0 in total (QC-passed reads + QC-failed reads)
1727 + 0 primary
429 + 0 secondary
382 + 0 supplementary
0 + 0 duplicates
0 + 0 primary duplicates
2538 + 0 mapped (100.00% : N/A)
1727 + 0 primary mapped (100.00% : N/A)
```

Decode Flag number

```
samtools flags 4
```

```
samtools flags 2048
```

```
samtools flags -l
```

```
samtools flags UNMAP,SECONDARY,SUPPLEMENTARY  
0x904 2308 UNMAP,SECONDARY,SUPPLEMENTARY
```

```
Apple > ~/Desktop/ONT/ONT_practice  
> samtools flags 4  
0x4 4 UNMAP  
  
Apple > ~/Desktop/ONT/ONT_practice  
> samtools flags 2048  
0x800 2048 SUPPLEMENTARY  
  
Apple > ~/Desktop/ONT/ONT_practice
```

```
About: Convert between textual and numeric flag representation  
Usage: samtools flags FLAGS...  
  
Each FLAGS argument is either an INT (in decimal/hexadecimal/octal) representing  
a combination of the following numeric flag values, or a comma-separated string  
NAME,...,NAME representing a combination of the following flag names:
```

0x1	1	PAIRED	paired-end / multiple-segment sequencing technology
0x2	2	PROPER_PAIR	each segment properly aligned according to aligner
0x4	4	UNMAP	segment unmapped
0x8	8	MUNMAP	next segment in the template unmapped
0x10	16	REVERSE	SEQ is reverse complemented
0x20	32	MREVERSE	SEQ of next segment in template is rev.complemented
0x40	64	READ1	the first segment in the template
0x80	128	READ2	the last segment in the template
0x100	256	SECONDARY	secondary alignment
0x200	512	QCFAIL	not passing quality controls or other filters
0x400	1024	DUP	PCR or optical duplicate
0x800	2048	SUPPLEMENTARY	supplementary alignment

Thống kê theo flag?

```
samtools view results/02_alignment/HG002.chr20.bam \
| cut -f2 \
| sort -n \
| uniq -c \
| xargs -n2 sh -c 'echo "$1\t$(samtools flags "$2")"' _
```

927	0x0	0	Primary mapped
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166	0x100	256	SECONDARY
263	0x110	272	REVERSE, SECONDARY
174	0x800	2048	SUPPLEMENTARY
208	0x810	2064	REVERSE, SUPPLEMENTARY

```
> samtools flagstats results/02_alignment/HG002.chr20.bam
2538 + 0 in total (QC-passed reads + QC-failed reads)
1727 + 0 primary
429 + 0 secondary
382 + 0 supplementary
0 + 0 duplicates
0 + 0 primary duplicates
2538 + 0 mapped (100.00% : N/A)
1727 + 0 primary mapped (100.00% : N/A)
```

Extract SAM/BAM theo Flag

```
# Count theo flags
```

```
samtools view -c -f 2048 results/02_alignment/HG002.chr20.bam
```

```
# Extract SAM/BAM theo Flag
```

```
samtools view -b -f 2048 results/02_alignment/HG002.chr20.bam \  
> results/02_alignment/supplementary.bam
```

```
-F: excluded
```

```
-f: included
```

```
# check flagstats
```

```
samtools flagstats results/02_alignment/supplementary.bam
```

CIGAR (Compact Idiosyncratic Gapped Alignment Report)

Op	BAM	Description	Consumes query	Consumes reference
M	0	alignment match (can be a sequence match or mismatch)	yes	yes
I	1	insertion to the reference	yes	no
D	2	deletion from the reference	no	yes
N	3	skipped region from the reference	no	yes
S	4	soft clipping (clipped sequences present in SEQ)	yes	no
H	5	hard clipping (clipped sequences NOT present in SEQ)	no	no
P	6	padding (silent deletion from padded reference)	no	no
=	7	sequence match	yes	yes
X	8	sequence mismatch	yes	yes

CIGAR (Compact Idiosyncratic Gapped Alignment Report)

1. Match / mismatch

CIGAR: 100M

Reference: ACGTACGTACGT...
Read: ACGTACGTACGT...
|||||||
100M

2. Insertion

CIGAR: 5M2I5M

Reference: ACGTA--CGTAC
Read: ACGTATTCGTAC
|||||II|

3. Deletion

CIGAR: 5M2D5M

Reference: ACGTATTCGTAC
Read: ACGTA--CGTAC
|||||DD|

4. Soft clipping

CIGAR: 3S7M

Reference: ACGTACG
Read: TTTACGTACG
SSS|||||

5. Hard clipping

CIGAR: 3H7M

Original read:
TTTACGTACG

SAM/BAM SEQ:
ACGTACG
|||||
7M

6. Exact match / mismatch

CIGAR: 3=1X3=

Reference: ACGTACG
Read: ACGAACG
===X===

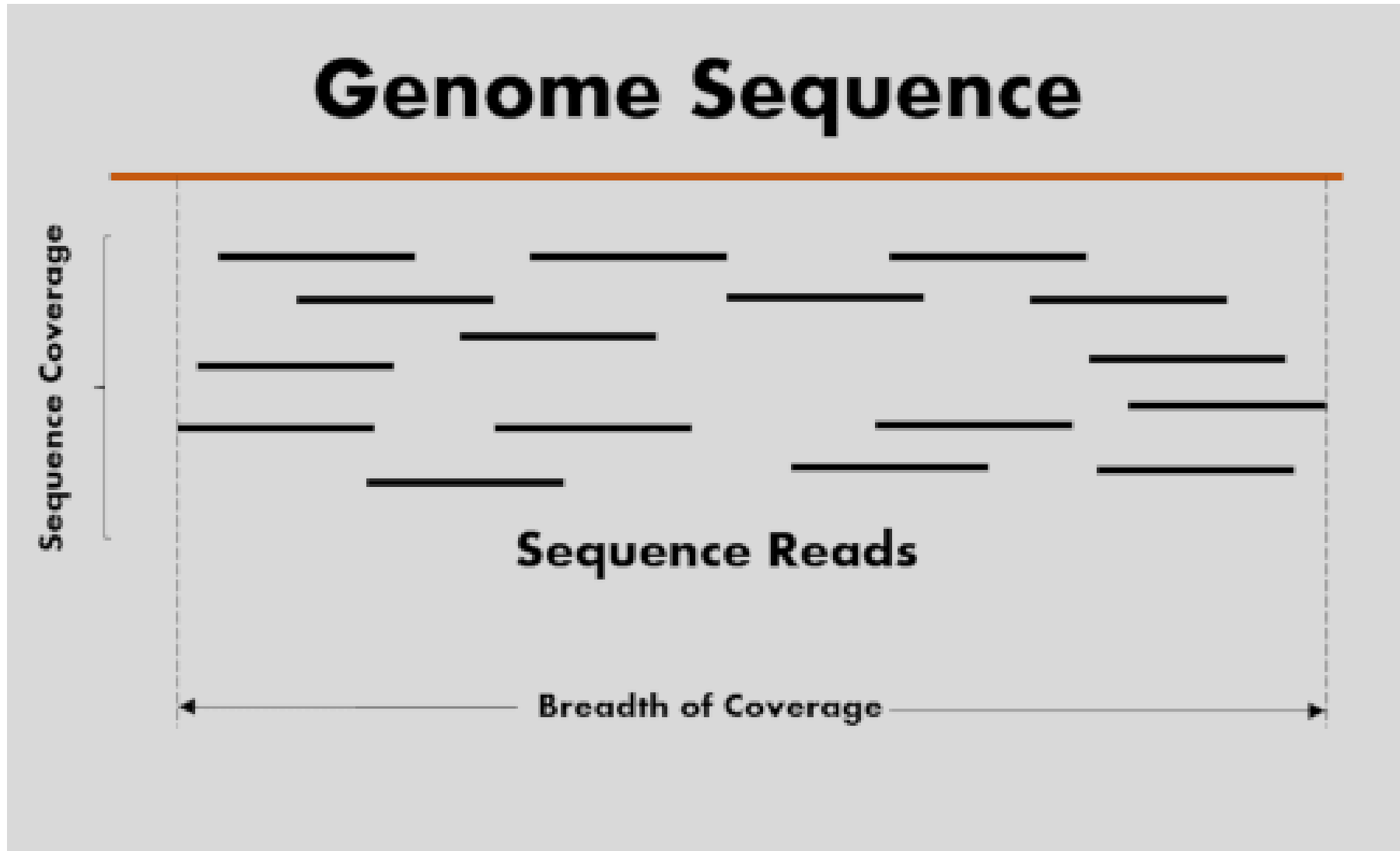
Mapping Quality

MAPQ = độ tin cậy rằng một alignment được đặt đúng vị trí trên reference genome

- MAPQ được gán cho **mỗi alignment record**, không phải cho từng base.
- MAPQ phản ánh **độ tin cậy của vị trí mapping**.
- Một read vẫn có thể được mapped nhưng có MAPQ thấp.
- MAPQ thực tế phụ thuộc vào thuật toán của aligner, nên không phải lúc nào cũng là xác suất được hiệu chuẩn tuyệt đối.

$$\text{MAPQ} = -10 \times \log_{10}(\text{P}(\text{mapping sai}))$$

Coverage



Depth Coverage

```
# Print depth coverage: Note: không in các vị trí depth = 0
samtools depth results/02_alignment/HG002.chr20.bam | head
```

```
# Print depth coverage của tất cả các vị trí
samtools depth -a results/02_alignment/HG002.chr20.bam | head
```

```
# tính depth coverage trung bình
samtools depth -a results/02_alignment/HG002.chr20.bam \
| awk '{sum += $3; n++} END {print sum/n}'
```

```
# tính nhanh bằng option coverage
samtools coverage results/02_alignment/HG002.chr20.bam | column -t
```

Samtools Coverage

```
# tính nhanh bằng option coverage
samtools coverage results/02_alignment/HG002.chr20.bam | column -t
```

Column	Giá trị	Ý nghĩa
rname	chr20	Reference/chromosome
startpos	1	Vị trí bắt đầu
endpos	64,444,167	Vị trí cuối / chiều dài chr20
numreads	2,109	Số alignment được chọn sau filtering
covbases	428,920	Số reference bases có depth ≥ 1
coverage	0.665568	% reference được cover
meandepth	0.444651	Mean depth trên toàn region
meanbaseq	36	Mean base quality của bases được xét
meanmapq	55.4	Mean mapping quality của selected alignments

Samtools Coverage theo region

```
# Export toàn bộ depth theo từng vị trí trong bed file
samtools depth -a -b \
config/target.chr20.bed \
results/02_alignment/HG002.chr20.bam > \
results/02_alignment/HG002.chr20.bed.cov.tsv
# view file
less results/02_alignment/HG002.chr20.bed.cov.tsv
```

chr20	63838	0
chr20	63839	0
chr20	63840	0
chr20	63841	52
chr20	63842	54
chr20	63843	54
chr20	63844	54
chr20	63845	54
chr20	63846	54
chr20	63847	54
chr20	63848	55

Depth of coverage

```
# Tính trung bình Depth of Coverage  
awk '{total+=$3; n++} END {print "Mean depth:", total/n "X"}'  
results/02_alignment/HG002.chr20.bed.cov.tsv
```

Mean Depth of Coverage: **99.704X**

Breath of coverage

```
# Tính trung bình Depth of Coverage
awk '{
    total++
    if ($3 >= 1) covered++
}
END {
    printf "Breadth of Coverage >=1x: %.2f%%\n", 100*covered/total
}' results/02_alignment/HG002.chr20.bed.cov.tsv
```

Breadth of Coverage >=1x: 99.99%

Breath of coverage: Nhiều threshold

```
# Tính trung bình Breath of Coverage
awk '{
    total++
    if ($3 >= 1) c1++
    if ($3 >= 10) c10++
    if ($3 >= 20) c20++
    if ($3 >= 30) c30++
}
END {
    printf "Mean/breadth positions: %d\n", total
    printf ">=1x : %.2f%%\n", 100*c1/total
    printf ">=10x: %.2f%%\n", 100*c10/total
    printf ">=20x: %.2f%%\n", 100*c20/total
    printf ">=30x: %.2f%%\n", 100*c30/total
}' results/02_alignment/HG002.chr20.bed.cov.tsv
```

Mean/breadth positions: 272200

>=1x : 99.99%

>=10x: 99.98%

>=20x: 99.97%

>=30x: 99.96%

PHẦN 2

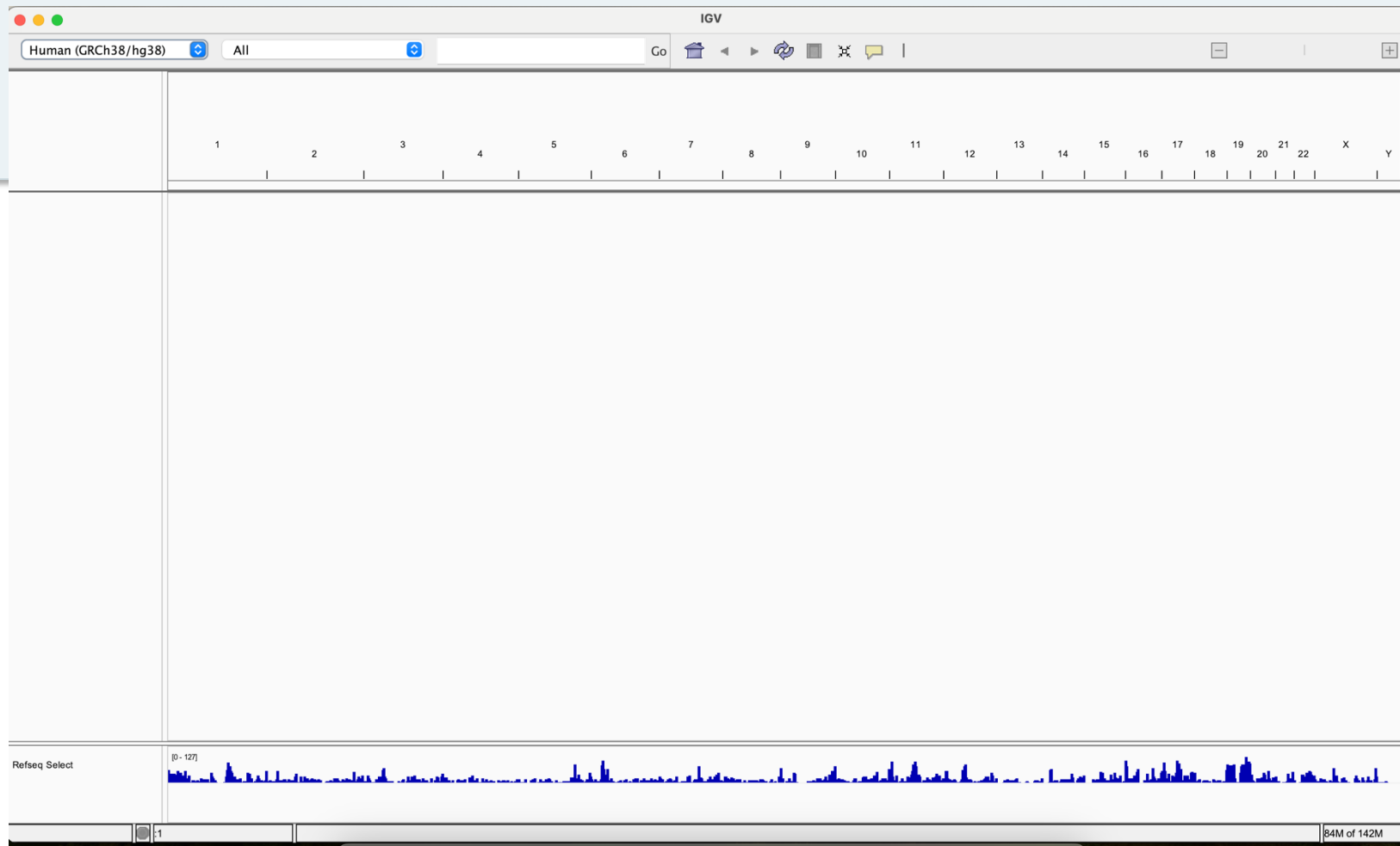
IGV



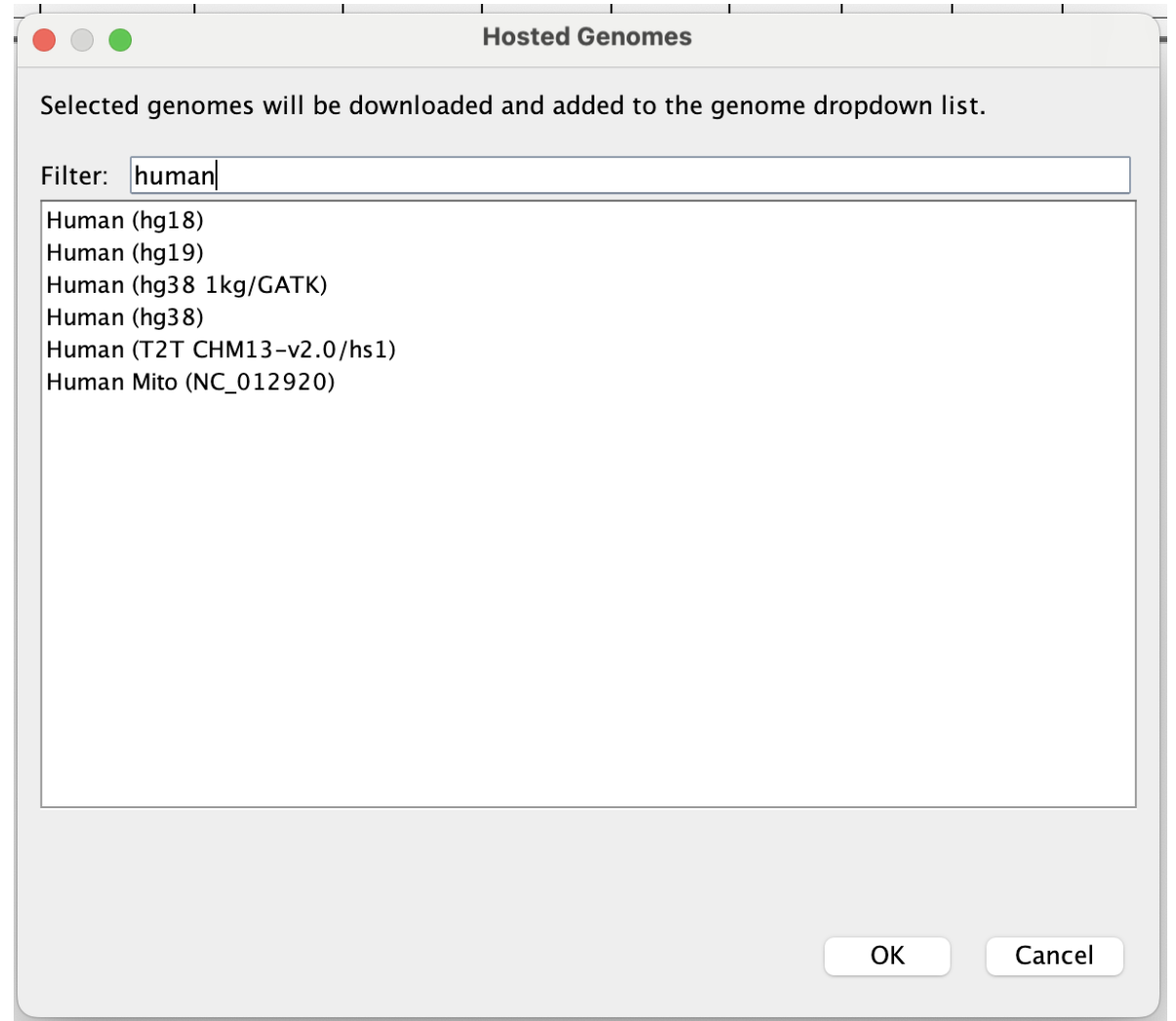
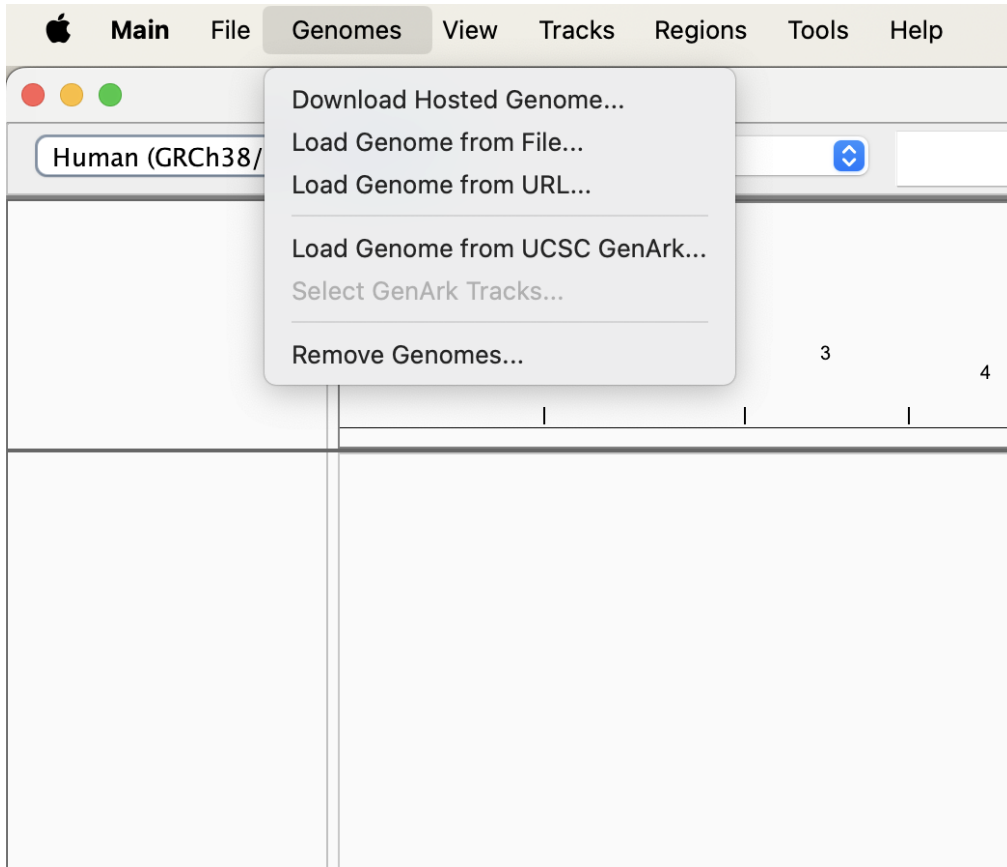
Open IGV

```
conda activate igv
```

```
igv
```

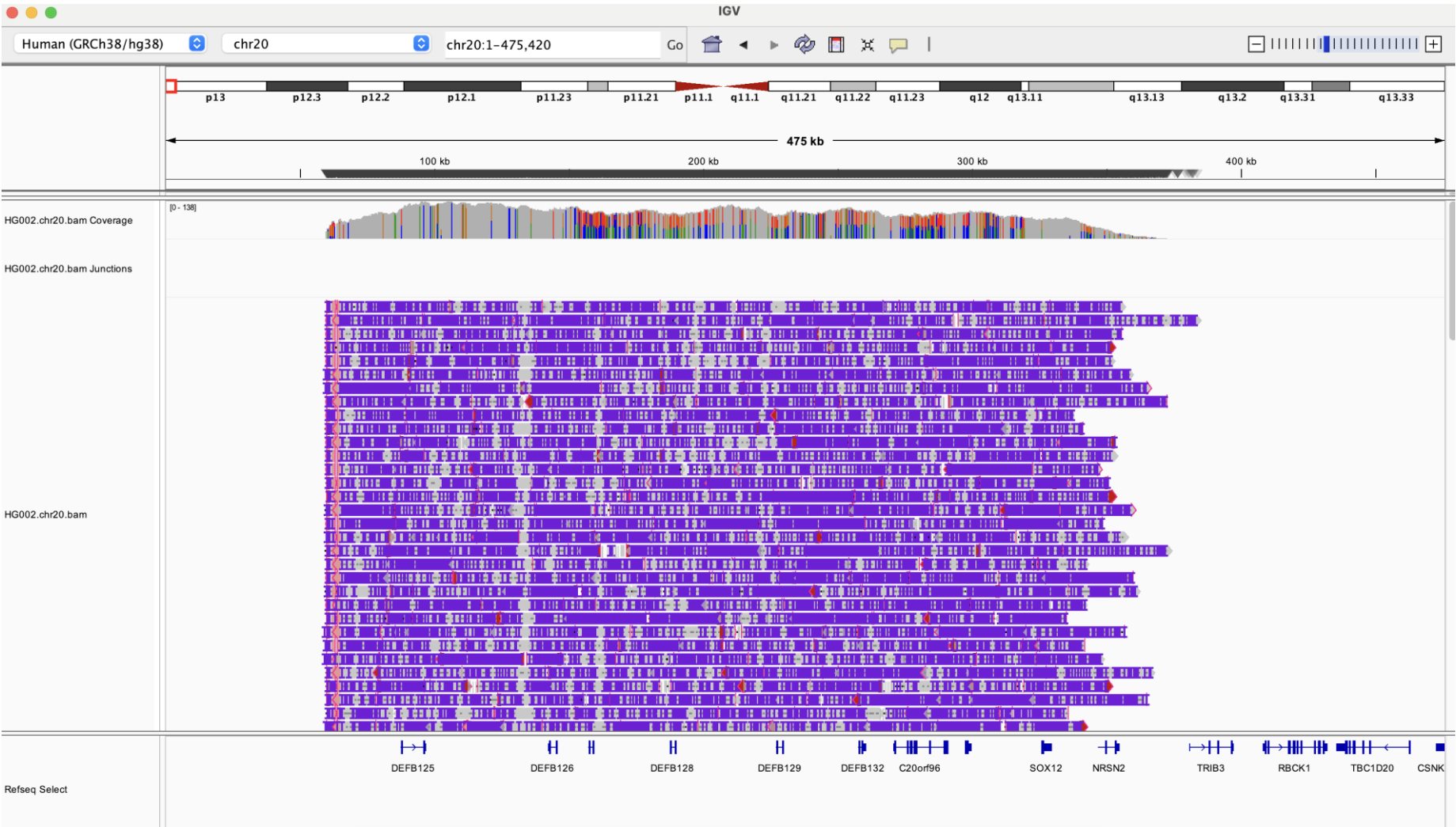


Load genome



Open BAM file

Nên thực hiện index BAM trước khi open



Bài tập luyện tập

Thực hiện các bước để alignment với chrM

- Extract reference
- Extract fastq
- QC
- Alignment